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# =====
# Dataset 1: Iris Dataset Analysis
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# 1. Load and Explore Data
data(iris)

# View structure
str(iris)

# See first rows
head(iris)

# =====
# 2. Basic Operations
# =====
avg_sepal_length <- mean(iris$Sepal.Length)
max_petal_length <- max(iris$Petal.Length)

print(avg_sepal_length)
print(max_petal_length)

# =====
# 3. Petal Size Classification
# =====
size_label <- c()

for (i in iris$Petal.Length) {
  if (i > 5) {
    size_label <- c(size_label, "Large")
  } else if (i > 3) {
    size_label <- c(size_label, "Medium")
  } else {
    size_label <- c(size_label, "Small")
  }
}

iris$Size <- factor(size_label)
head(iris)

# =====
# 4. Visualization
# =====

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# Scatter plot
plot(iris$Sepal.Length, iris$Petal.Length,
     col = iris$Species,
     main = "Sepal vs Petal Length",
     xlab = "Sepal Length",
     ylab = "Petal Length")

legend("topleft",
      legend = levels(iris$Species),
      col     = 1:3,
      pch     = 1)

# Boxplot
boxplot(Petal.Length ~ Species, data = iris,
        col = c("red", "green", "blue"),
        main = "Petal Length by Species")
```